



Tutorial 1 Microfluidics (Claus-Dieter Ohl)

Please prepare the answers for the following questions for our tutorial on 28.5.2026.

$$\begin{aligned} u &= a x \\ v &= -a y \quad \text{where } a = \text{const} > 0 \end{aligned}$$

1. Streamlines

Given the following flow field:

$$\begin{aligned} u &= a x \\ v &= -a y \quad \text{where } a = \text{const} > 0 \end{aligned}$$

- Find an equation for the streamline that passes through a point (x_0, y_0) . In which direction moves a fluid particle? If it originates from a location $y_0 > 0$ does the particle pass through the horizontal line $y = 0$. Where is the stagnation point? What kind of flow could that be?
- Plot the flow field (arrow or `plt.quiver` plot for $x \geq 0$). For $x_0 = 1$ and $y_0 = 1$, plot a streamline passing through this point. Then, plot through nearby points a few more streamlines to illustrate the flow field.

2. Index Notation

Use the index notation and show that these are equalities:

a.

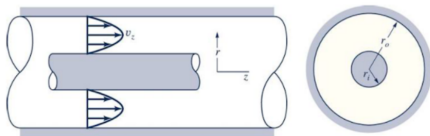
$$\sum_{i=1}^3 \frac{\partial}{\partial x_i} (p \delta_{ij}) = (\nabla p)_j$$

b.

$$\nabla \cdot (\rho \vec{u}) = (\nabla \rho) \cdot \vec{u} + \rho \nabla \cdot \vec{u}$$

3. Navier Stokes Equation

Repeat the Navier Stokes derivation for a tube now for 2 concentric tubes with radii r_i and r_o .



Show that the flow velocity in z -direction is

$$u_z(r) = \frac{1}{4\mu} \frac{dp}{dz} \left(r^2 - r_o^2 + \frac{r_i^2 - r_o^2}{\ln \frac{r_o}{r_i}} \ln \left[\frac{r}{r_i} \right] \right)$$

4. Flow Simulations

Study the program for the Navier Stokes solver provided in the course `13a_Navier Stokes example.ipynb`.

- Write down the boundary conditions (b.c.) for the velocity $\vec{u}$ and pressure for a flow in a 2d-tube driven by a pressure gradient. You need one b.c. for each boundary, i.e. the walls and the inlet/outlet, and component of the velocity. The pressure at the walls are determined by the incompressibility condition. For help, look into the code.
- Do the same for a flow in a 2-tube driven by the upper wall. Again, check your answer with the boundary conditions stated in the code.
- Conduct simulations for pressure driven and wall driven flows. Try to obtain a parabolic profile for the pressure driven flow and a linear profile for the wall driven flow. Discuss the results. What is the effect of the time step Δt of integration.